

PLASTIC-CRAFT

Polyethylene

LDPE & HDPE Comparison | See separate sheet for UHMW-PE

845-358-3010
ecom@plastic-craft.com
www.plastic-craft.com

The polyethylene family offers the best chemical resistance of any commodity thermoplastic at the lowest cost. LDPE provides maximum flexibility and impact toughness; HDPE adds stiffness, tensile strength, and higher temperature resistance. Both are FDA/USDA compliant and non-toxic.

STOCK AVAILABILITY

Plastic-Craft maintains extensive inventory across a wide range of standard and specialty sizes in sheets, rods, tubes, and custom profiles. Most orders ship from stock — contact us for immediate availability.

WHEN TO CHOOSE

- LDPE** Gaskets, flexible liners, squeeze bottles, soft bumpers, packaging
- HDPE** Tanks, cutting boards, wear parts, marine, structural components

KEY HIGHLIGHTS

- Outstanding chemical resistance
- Very low moisture absorption
- Lightest thermoplastics
- Non-toxic & non-staining
- FDA / USDA / NSF / 3A-Dairy
- ✓ Both grades float in water (SG < 1.0)

GRADES AVAILABLE

- Natural (white / translucent)
- Black (UV-stabilized)
- FDA-compliant
- Marine Board (HDPE)
- Cutting Board (HDPE)
- Colors available

MATERIAL PROPERTIES — LDPE VS HDPE COMPARISON

PROPERTY	ASTM	LDPE	HDPE	UNITS
PHYSICAL				
Specific Gravity	D792	0.92	0.96	—
Water Absorption, 24h	D570	< 0.01	< 0.01	%
MECHANICAL				
Tensile Strength, Ultimate	D638	1,400	4,000	psi
Tensile Modulus	D638	30,000	200,000	psi
Elongation at Break	D638	500	500	%
Flexural Modulus	D790	29,000	200,000	psi
Compressive Strength (10%)	D695	1,400	3,500	psi
Impact, Izod Notched	D256	No Break	1.5	ft-lb/in
Hardness, Shore D	D2240	44 – 50	65 – 70	Shore D
FRICTION / WEAR				
Coefficient of Friction (Dynamic)	D1894	0.25	0.20	vs Steel
Coefficient of Friction (Static)	D1894	0.30	0.25	vs Steel
Abrasion Resistance (Taber)	D1044	25	15	mg/1000 cyc
THERMAL				
Heat Deflection @ 264 psi	D648	105	150	°F
Heat Deflection @ 66 psi	D648	120	180	°F
Melting Point	D3418	230	266	°F
CTE (Linear)	D696	8.5 × 10 ⁻⁵	7.0 × 10 ⁻⁵	in/in/°F
Thermal Conductivity	C177	2.2	3.0	BTU-in/hr-ft²-°F
Max Continuous Service	—	160	180	°F
ELECTRICAL				
Volume Resistivity	D257	> 10 ¹⁶	> 10 ¹⁶	ohm-cm
Dielectric Strength	D149	460	500	V/mil
Dielectric Constant, 1MHz	D150	2.3	2.3	—
FLAMMABILITY				
UL Flammability Rating	UL 94	HB	HB	—
Weathering / UV	G154	Fair	Fair	UV-stab. grades avail.

PROCESSING / FABRICATION NOTES: Both grades machine well with standard tooling | HDPE: weldable by hot plate, hot gas & extrusion welding; thermoformable | LDPE: highly flexible, excellent for gaskets & linings | Neither bonds well with adhesives — use mechanical fastening

CHEMICAL RESISTANCE: Excellent — both grades resist most acids, bases, alcohols, and aqueous solutions. **AVOID:** strong oxidizing acids, aromatic & chlorinated hydrocarbons at elevated temperatures. HDPE stress-crack resistant grades available for aggressive chemical service. **NOTE:** For UHMW-PE properties (highest impact, lowest friction, best abrasion resistance), see our dedicated UHMW spec sheet.

Typical values; not to be used as specifications. Actual values may vary by grade, manufacturer, and processing conditions. Contact Plastic-Craft for grade-specific data.